

Zahid Jabbar Khan

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Education

Birla Institute of Technology, Mesra,,Ranchi

B.Tech – Electronic and Communication Engineering (6.98 CGPA)

Dec 2021 - May 2025

Warren Academy School, Jaipur

12th PCM –Percentage: 77%

March 2020 – April 2021

Experience

R&D Intern – Genus Power Infrastructure Ltd. | [Internship Offer Letter](#) | [Internship Certificate](#)

June 2024 - August 2024

- Introduced **bsdiff** and **bspatch** into projects to simplify the process of creating and applying binary patches. Leveraged the power of [Xdelta3](#) for **delta encoding**, expanding support to multiple file types.
- Achieved advanced integration of bsdiff and Xdelta3 within the [File Zipper Project](#), creating a hybrid solution that combines compression and patching.
- Enhanced the **File Zipper** project by incorporating binary diffing and patching techniques, ensuring a user-friendly tool for efficient file management. [Project - GitHub Link](#)

Frontend Development Intern – BrandNav | [Internship Offer Letter](#) | [Internship Certificate](#)

Jan 2025 - Feb 2025

- Helped in developing [BrandNav](#) homepage with optimized responsive UI components using **React.js**, **Next.js**, and **Astro.js** ensuring seamless user experiences and also leveraged **Tailwind CSS** for efficient styling, improving UI consistency and maintainability.
- Implemented **TypeScript** to enhance code reliability and scalability and built a [Email Verifier Prototype](#) with dynamic content management and an intuitive UI.
- Developed [AirVault](#), a web-based application, consisting of a homepage and a [dashboard](#) focusing on secure document storage and accessibility. [Project - GitHub Link](#)

Backend Development Intern – CozCore Technology

March 2025 - Present

- Developed and optimized blockchain-based applications** using **Node.js**, **PHP**, **Vue.js**, and **Tailwind CSS**, ensuring high performance and security.
- Designed and tested RESTful APIs** with **Postman** and **Node.js**, improving system efficiency and backend communication.
- Integrated smart contract functionality** into web applications, enhancing decentralized solutions for clients.

Skills

Libraries/Frameworks:	Node.js, React.js, Express.js, JWT, Passport.js
Programming Languages:	CPP, C, Python, HTML, CSS, Javascript, SQL
Tools / Platforms:	VS-Code, Git, GitHub, NI LabView, MySQL, MongoDB, Leetcode,
Technical Skills:	Data Structures, Computer Networks, DBMS, OOPS, Operating System

Projects

SmartZip – Advanced Compression Tool | July 2024 - August 2024

- Engineered a high-performance **file compression tool** in **C++** using **Huffman coding**, achieving up to **30%** reduction in file size while ensuring **100%** lossless decompression.
 - Integrated advanced file comparison using [bsdiff4](#)(**C and Python**), generating **25% smaller patch files** than standard tools, reducing storage costs significantly.
 - Designed **automated testing scripts** to validate performance and correctness, streamlining the verification of compression.
- GitHub Link : <https://github.com/Zahidmohd/File-Zipper>

Integrated Hospital Management and ECG Monitoring System | Dec 2023 - Sep 2024

- Designed a [hardware-software solution](#) integrating **ECG signal acquisition** and hospital management functionalities, enhancing **real-time monitoring and data management**.
- Utilized **LabVIEW** to process ECG signals acquired through a **DAQ device**, extracting features like **PQRST intervals** and **diagnosing cardiac abnormalities**.
- Developed a [dashboard](#) using **React.js** to visualize live ECG waveforms, enabling real-time monitoring and engineered a backend with **Node.js** and **MySQL** to store and retrieve patient ECG data, ensuring seamless integration with the hospital database.

Achievements

- Presented a Research Paper** titled [Real-Time ECG Monitoring and Abnormality Detection with LabVIEW](#) in National Conference on Recent Trends in Space Technology (RTST-2024) by **ISRO – MNIT RAC-S ,MNIT JAIPUR**.
- Cirquik College Hackathon (Intracollege)** – IETE and CSI : **Stood 1st** among **33 teams** participating in it. Worked on an ESG(Environmental ,Social, and Governance) problem statement that aims to **predict agricultural and farmer based emissions** using web applications and leveraging machine learning models. [Certificate](#)
- Leetcode platform** – Solved over **190+ problems** and **won 2 badges**. [Profile](#)